

Call Center Performance Analysis

Portfolio Project Case Study | Power BI | January 2026

1. Project Overview

This project analyzes call-center operational performance using Power BI, focusing on service level, staffing coverage, intraday demand, adherence, and SLA breaches. The objective was to move beyond high-level reporting and identify when and where workforce coverage gaps were affecting customer service.

The final report contains five interactive pages: Executive Overview; Workforce & Staffing Analysis; Interval & Root-Cause Analysis; Adherence & Worst Performance Observations; and Business Insights & Recommended Actions.

2. Business Problem

Call-center service performance can deteriorate when staffing does not align with intraday call demand. Aggregate staffing totals may hide short periods of understaffing, while good adherence alone does not guarantee sufficient coverage. This analysis therefore examines demand, required staffing, scheduled staffing, service level and adherence at both queue and interval level.

3. Project Objectives

- Evaluate service-level performance against the 85% target.
- Identify SLA breach intervals and the queues most affected.
- Assess staffing gaps and required versus scheduled staffing.
- Examine intraday patterns in demand and service performance.
- Assess adherence alongside staffing coverage.
- Identify performance hotspots and translate findings into operational recommendations.

4. Dataset

The analysis uses a call-center operational dataset containing date, day, interval, queue, call-volume, service-level, staffing and adherence information. The data was prepared and validated in Power Query before analysis in Power BI.

Field	Description
Date	Date of the operational record
Day	Day of the week
Interval	Operating time interval
Queue	Service queue
Calls Offered	Calls presented to the queue
Calls Answered	Calls answered
Calls Abandoned	Calls abandoned
Service Level %	Percentage of calls meeting the SLA
Staffing Gap	Difference between required and scheduled staffing
Adherence %	Adherence to scheduled activity

5. Tools & Techniques

- Power BI Desktop — dashboard development, modeling and interactive reporting.
- Power Query — data preparation, type validation and quality checks.
- DAX — KPI measures and staffing-gap calculations.
- Power BI visualizations — KPI cards, line charts, bar charts, scatter plots and analytical tables.
- Report-page tooltips — contextual interval-level detail.

6. Methodology

- Data preparation: validated date and interval fields and checked record consistency.
- Performance validation: reviewed service level, staffing gaps and adherence.
- KPI development: created measures for calls offered, service level, abandon rate, AHT, SLA breaches and cumulative staffing gap.
- Intraday analysis: compared service performance, demand and staffing requirements across intervals.
- Queue analysis: compared service level, breach counts and staffing gaps by queue.
- Exception analysis: identified the lowest service-level and staffing observations.
- Business interpretation: translated patterns into workforce-management recommendations.

7. Key Performance Indicators

KPI	Result	Interpretation
Total Calls Offered	96,506	Overall demand
Average Service Level	87.4%	Above 85% target
SLA Target	85%	Performance benchmark
Abandon Rate	2.5%	Share of calls abandoned
Average AHT	426.9 seconds	Average handling time
SLA Breach Intervals	1,437	Intervals below target
Cumulative Staffing Gap	311	Net staffing-gap indicator

8. Dashboard Structure

- Page 1 — Executive Overview: management-level view of calls, service level, abandon rate, AHT, staffing gap and trends.
- Page 2 — Workforce & Staffing Analysis: daily staffing gaps, staffing versus service level, required versus scheduled staffing and queue gaps.
- Page 3 — Interval & Root-Cause Analysis: intraday service level, staffing, demand and interval gaps.
- Page 4 — Adherence & Worst Performance Observations: adherence patterns and lowest-performing observations.
- Page 5 — Business Insights & Recommended Actions: business interpretation and operational recommendations.

9. Key Findings

- Overall SLA was 87.4%, exceeding the 85% target; however, 1,437 intervals breached the SLA target.
- Technical Support recorded the highest number of SLA breach intervals and was among the queues with the largest cumulative staffing gaps.
- The lowest-SLA observations frequently occurred when staffing was below required staffing, indicating that staffing shortages are an important contributor to severe SLA deterioration.
- Service-level performance generally improved as staffing coverage increased, although staffing was not the only factor influencing SLA.
- Staffing requirements varied considerably throughout the day, creating opportunities for more precise intraday workforce alignment.
- Several low-SLA observations were concentrated around specific intervals, including recurring observations around 2:30 PM.
- Average adherence remained around 90%, suggesting that SLA challenges cannot be explained by adherence alone.

10. Recommendations

- Rebalance intraday staffing: adjust coverage around periods where demand and staffing requirements increase.
- Investigate Technical Support: review staffing capacity, AHT, call complexity and scheduling patterns.

- Investigate recurring SLA hotspots: use the worst-observation tables to identify repeated interval/queue combinations.
- Monitor adherence alongside staffing: avoid using adherence as the only workforce-performance indicator.
- Forecast and schedule at interval level: use historical call volume, AHT and required staffing to build more precise schedules.

11. Analyst Takeaways

Aggregate staffing figures do not tell the full story. A positive or improving overall staffing position can coexist with short periods of understaffing and SLA breaches. Effective workforce analysis therefore requires looking at when, where, and how much staffing coverage deviates from demand.

The project also reinforced the importance of distinguishing correlation from causation. Staffing coverage provides an important operational signal, but AHT, queue mix, call complexity and other operational factors can also influence performance. Combining multiple dimensions produces a stronger business diagnosis.

12. Portfolio Value

This project demonstrates practical ability in data preparation, KPI development, DAX-based analysis, Power BI dashboard design, intraday workforce analysis, exception reporting and business storytelling. The output is designed to support both technical review and business-oriented discussion during a data analyst or business analyst portfolio review.

Project status: Completed Power BI dashboard with five analytical pages, interactive navigation, report-page tooltips and a dedicated business-insights page.